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Sr. No. of Question Paper : 2065

GC-3

Your Roll No.....

Unique Paper Code : 32351302

Name of the Paper : C6 Group Theory I

Name of the Course : B.Sc. (Hons) Mathematics – CBCS

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on the receipt of this question paper.
2. Attempt any **two** parts from each question.
3. All questions are compulsory.

1. (a) Describe the elements of the dihedral group D_3 and show that they form a group under composition of mappings. (6)
(b) Prove that if H and K are subgroups of G , then so is $H \cap K$. Is $H \cup K$ a subgroup of G ? Justify your answer. (6)
(c) Define Centralizer $C(G)$ of a group G . Is $C(G)$ Abelian? Justify your answer. (6)
2. (a) Let $G = \langle a \rangle$ be a cyclic group of order n . Show that $G = \langle a^k \rangle$ if and only if $\gcd(k, n) = 1$. (6)

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(b) List the subgroups of $\mathbb{Z}_{30}$ and their generators. (6)

(c) (i) Let G be an Abelian group with identity e . Prove that

$$H = \{x \in G \mid x^2 = e\} \text{ is a subgroup of } G. \quad (3)$$

(ii) Let a and b be elements of a group. If $|a| = 10$ and $|b| = 21$, show

$$\text{that } \langle a \rangle \cap \langle b \rangle = \{e\}. \quad (3)$$

3. (a) Prove that every permutation on a finite set can be written as a cycle or product of disjoint cycles. (6)

(b) Prove that the set of even permutations in S_n forms a subgroup of S_n . (6)

(c) Let $\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 1 & 3 & 5 & 4 & 7 & 6 & 8 \end{bmatrix}$ and $\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 3 & 8 & 7 & 6 & 5 & 2 & 4 \end{bmatrix}$

Write α and β as

(i) products of disjoint cycles.

(ii) products of 2-cycles. (6)

4. (a) Prove that any finite cyclic group of order n is isomorphic to $\mathbb{Z}_n$ and any infinite cyclic group is isomorphic to $\mathbb{Z}$. (6½)

(b) Let $\bar{G}$ denote the left regular representation of the group G (as defined in Cayley's Theorem). Calculate $\overline{U(12)}$. (6½)

- (e) Give statements only, of Lagrange's Theorem and its converse. Is the converse true? Justify your answer. (6½)

5. (a) Let H and K be subgroups of a finite group G . Prove that

$$|HK| = \frac{|H||K|}{|H \cap K|} \quad (6\frac{1}{2})$$

- (b) Let G be a finite Abelian group and let p be a prime number that divides the order of G . Prove that G has an element of order p . (6½)

- (c) Let ϕ be a homomorphism from a group G to a group $\bar{G}$ and let H be a subgroup of G . Prove that

(i) if H is cyclic, then $\phi(H)$ is cyclic. (2)

(ii) if H is Abelian, then $\phi(H)$ is Abelian. (2)

(iii) if H is normal in G , then $\phi(H)$ is normal in $\phi(G)$. (2½)

6. (a) State and prove The Second Isomorphism Theorem. (6½)

- (b) Let G be a subgroup of some dihedral group. For each x in G , define

$$\phi(x) = \begin{cases} +1 & \text{if } x \text{ is a rotation} \\ -1 & \text{if } x \text{ is a reflection} \end{cases}$$

Prove that ϕ is a homomorphism from G to the multiplicative group $\{+1, -1\}$. What is the kernel of ϕ ? (6½)

(c) Determine all homomorphisms from $\mathbb{Z}_{12}$ to $\mathbb{Z}_{30}$. (6½)